Definition (The category $\langle\!\langle \mathbf{Pos} \rangle\!\rangle$)
The category $\langle\!\langle \mathbf{Pos} \rangle\!\rangle$ is the subcategory of **Pos** where the objects are tuple posets, posets of the form

$$\langle\!\langle \mathbf{P}_1, \ldots, \mathbf{P}_n \rangle\!\rangle = \left\langle \langle\!\langle \mathbf{P}_1, \ldots, \mathbf{P}_n \rangle\!\rangle, \preceq_{\langle\!\langle \mathbf{P}_1, \ldots, \mathbf{P}_n \rangle\!\rangle} \right\rangle.$$